

# OmniLaTeX: A Universal Document Template

Jane Doe, John Smith  
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Introduction

OmniLaTeX is a modular, universal  $\LaTeX$  template built on KOMA-Script. It supports multiple document types including theses, articles, books, reports, and now posters — all from a single unified codebase.

Key design goals:

- Write once, render in any format
- Consistent typography across all document types
- Minimal boilerplate with sensible defaults

Methods

We built OmniLaTeX on top of KOMA-Script, leveraging its modular architecture to create a pluggable doctype system. Each document type is defined as a self-contained profile that configures:

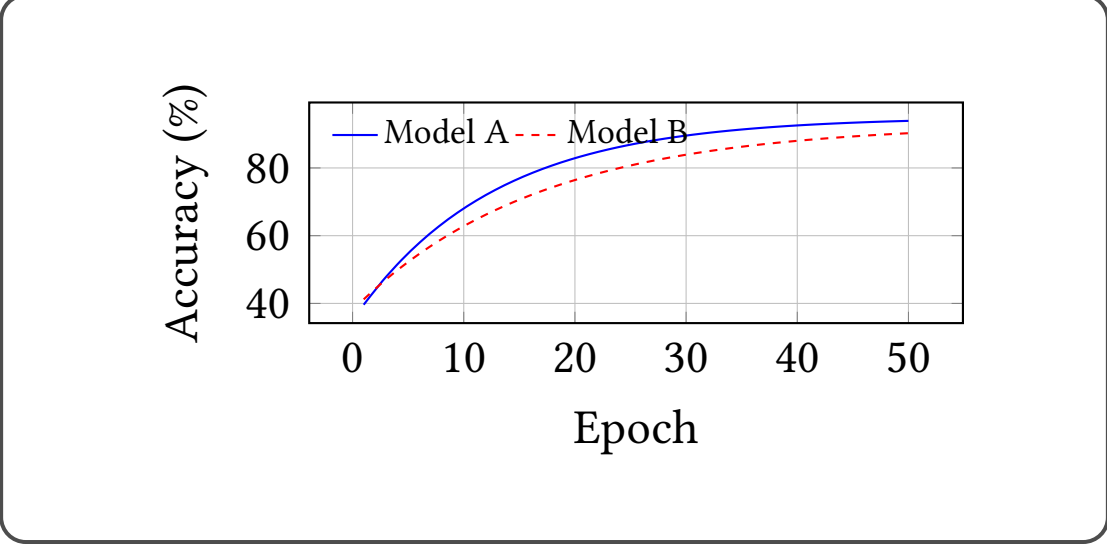
- Page layout and typography
- Section numbering depth
- Title page style
- Custom commands

All profiles share the same core infrastructure including bibliography management, glossaries, and code listings.

Results

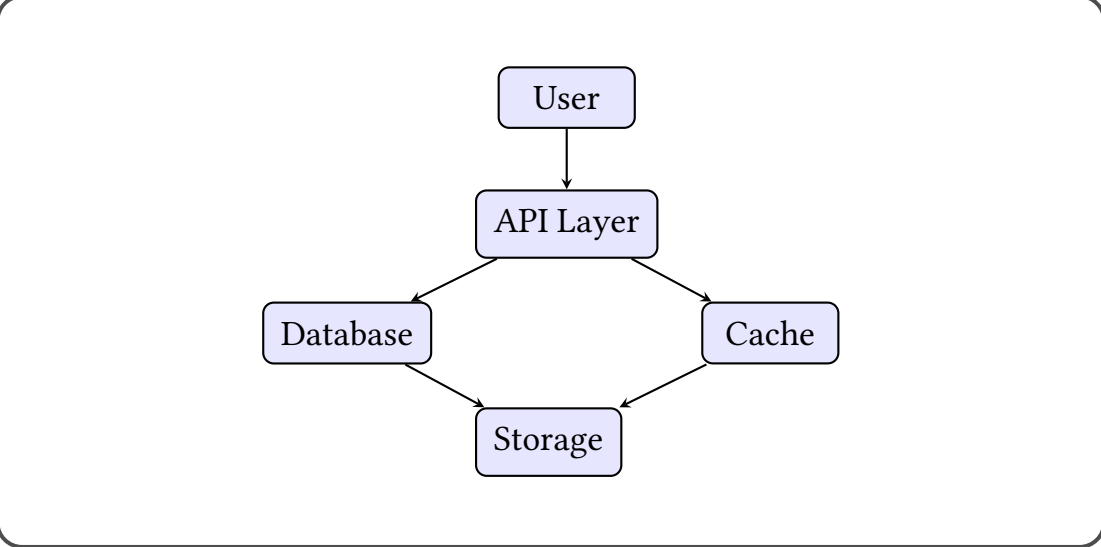
OmniLaTeX currently supports 16+ document types with full LuaLaTeX compilation. The template achieves:

- Reproducible builds via Docker and Nix
- Automatic bibliography and glossary generation
- Integrated code listings with syntax highlighting
- Multi-language support via polyglossia



| Epoch | Model A Accuracy (%) | Model B Accuracy (%) |
|-------|----------------------|----------------------|
| 0     | 40                   | 40                   |
| 10    | 65                   | 60                   |
| 20    | 80                   | 75                   |
| 30    | 88                   | 82                   |
| 40    | 90                   | 84                   |
| 50    | 90                   | 85                   |

Architecture Overview



```
graph TD; User --> API[API Layer]; API --> Database; API --> Cache; Database --> Storage; Cache --> Storage;
```

References

1. KOMA-Script Documentation, [ctan.org/pkg/koma-script](https://ctan.org/pkg/koma-script)
2. LaTeX Project Team, *The  $\LaTeX$  Companion*, 2023
3. OmniLaTeX Repository, [github.com/WyattAu/OmniLaTeX-template](https://github.com/WyattAu/OmniLaTeX-template)